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MAJOR PROJECT REPORT

on

DEEP LEARNING AND COMPUTER VISION-BASED PLANT DISEASE DETECTION SYSTEM WITH DOMAIN ADAPTATION

Submitted in partial fulfillment of the requirements for the award of the degree of

BACHELOR OF TECHNOLOGY

in

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

by

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(An Institute of National Importance under Ministry of Education, Govt. of India)

MAY, 2026



भारतीय सूचना प्रौद्योगिकी संस्थान भागलपुर
INDIAN INSTITUTE OF INFORMATION TECHNOLOGY BHAGALPUR
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DECLARATION

The work presented in this B.Tech Major Project, titled “Deep Learning and Computer Vision-Based Plant Disease Detection System with Domain Adaptation” is original. It was conducted independently by the students in the Department of Computer Science and Engineering at Indian Institute of Information Technology Bhagalpur, under the guidance of Dr. Thejaswini M.

No part of this work has been submitted to acquire any other Degree, Diploma, Fellowship, or similar titles at any other university or institution.

Date: May 2026

Place: IIIT Bhagalpur

Aman Kumar Bind (2201086CS)

Ankit Kumar Patel (2201116CS)

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CERTIFICATE

This is to certify that the project entitled “Deep Learning and Computer Vision-Based Plant Disease Detection System with Domain Adaptation” is carried out by:

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who are B.Tech students of Computer Science and Engineering at IIIT Bhagalpur, under my supervision and guidance. This project has been submitted in partial fulfillment of the requirements for the award of the “Bachelor of Technology” degree.

To the best of my knowledge, no part of this project report has been submitted for the award of any other degree or diploma.

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(Supervisor)

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Assistant Professor & Head, Dept. of CSE
(Head of Department)

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May 2026, IIIT Bhagalpur

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ABSTRACT

Global food security relies heavily on agricultural productivity, but plant diseases constantly reduce crop yields. Annual crop losses in India alone hit between 15% and 20% due to pathologies that go undetected or get diagnosed too late. Agronomists usually rely on physical inspections for diagnosis. This manual process is slow, prone to mistakes, and hard to access for farmers in remote areas. The proposed project offers a solution by combining a specialized Convolutional Neural Network (CNN) called ‘AgriNet’ with a Retrieval-Augmented Generation (RAG) chat engine.

Most agricultural computer vision models are trained using controlled laboratory images. This creates a large domain gap, causing them to fail frequently when tested in messy, real-world farm environments with varied lighting and backgrounds. The current work solves this problem through Mixed-Domain Dataset Fusion. Initially, the base model was trained on 54,305 lab images from the PlantVillage dataset, reaching a validation accuracy of 98.75%. To fix the domain gap, the system integrates 2,231 field images from the PlantDoc dataset. Mapping 27 real-world categories into the standard 38-class structure and fine-tuning the network on both domains resulted in a mixed-domain validation accuracy of 96.14%.

A custom, lightweight CNN backbone was developed from scratch to replace heavy pre-trained networks. It uses Inverted Residual Blocks and Depthwise Separable Convolutions. This specific design lowers the parameter count and speeds up inference, making it suitable for hardware with limited resources. Gradient-weighted Class Activation Mapping (Grad-CAM) adds an explainability layer by highlighting the exact diseased areas on the leaf.

Alongside the classification features, a local knowledge-grounded RAG engine was built. Authoritative agricultural manuals were split into vector embeddings and saved in a local FAISS database. When users submit queries, the system pulls relevant treatment context and synthesizes an answer via the Groq API (using Llama-3.1-8b-instant). This process prevents the generation of false information and ensures all treatment advice is verified. A FastAPI backend handles the logic, connected to a Next.js frontend with glassmorphic visuals and downloadable PDF reports.

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Chapter 1

INTRODUCTION

1.1 General

Agriculture represents the cornerstone of the global economy, supplying vital sustenance, raw materials, and livelihoods to billions worldwide. In developing nations like India, the agricultural sector employs over half the national workforce and contributes significantly to the country's Gross Domestic Product (GDP). However, agricultural yield remains constantly threatened by biological pathogens, environmental stressors, and invasive pests. Pathological infections—predominantly fungal, bacterial, and viral diseases—inflict catastrophic crop losses, resulting in an annual reduction of approximately 15% to 20% in India's total agricultural output.

Early identification of crop diseases is imperative to mitigate these losses. Traditionally, farmers diagnose plant pathologies through visual inspection of leaf symptoms. However, visual diagnostics are highly subjective, error-prone, and labor-intensive. Moreover, accessing professional plant pathologists remains difficult in remote, rural farming communities. Consequently, inaccurate diagnoses often prompt farmer to apply incorrect chemical pesticides, which not only escalates production cost but also causes soil degradation and environmental toxicity.

Automating disease detection using computer vision and deep learning techniques offers a scalable, objective solution. Convolutional Neural Networks (CNNs) have shown state-of-the-art performance in image classification. Nonetheless, most academic models are developed using pristine, laboratory-captured datasets, leading to a performance collapse when deployed in natural, unstructured field environments. To address this domain gap, we present a unified system featuring: (i) a custom, lightweight CNN backbone called ‘AgriNet’; (ii) a mixed-domain training pipeline for domain adaptation; (iii) Explainable AI (XAI) overlays via Grad-CAM; and (iv) a Retrieval-Augmented Generation (RAG) assistant that provides context-grounded, verified agricultural advice.

1.2 Literature Survey

To establish the research context, a thorough survey of literature spanning traditional machine learning, deep learning architectures, domain adaptation, and retrieval systems in agriculture was performed.

1.2.1 Traditional Classifiers vs. Deep Learning Models

Prior to the dominance of Deep Learning, computer vision in agriculture relied on handcrafted features paired with classical classifiers. Sarunyagate et al. [1] and other early researchers proposed image segmentation techniques to isolate lesion areas on leaves, followed by feature extraction. These features were classified using Support Vector Machines (SVM), Random Forests, or K-Nearest Neighbors (KNN). A major limitation of these systems was their dependency on manual feature selection, which is highly sensitive to changes in leaf color, size, and lighting conditions.

With the breakthrough of AlexNet and subsequent models like ResNet and VGG16 [3], the paradigm shifted to Deep Learning. CNNs automate the feature extraction process by learning hierarchical representations directly from raw image pixels. However, standard deep learning models like VGG16 are computationally heavy, containing over 138 million parameters, making them unsuitable for real-time mobile deployment or execution on low-power devices in fields.

1.2.2 Domain Adaptation in Agricultural Computer Vision

The availability of standardized datasets has accelerated deep learning research. The most prominent dataset is PlantVillage [2], introduced by Hughes and Salathé, which contains over 54,000 images captured in controlled laboratory settings with uniform grey backgrounds. Models trained on PlantVillage regularly report classification accuracies exceeding 98%. However, Mohanty et al. [10] demonstrated that when these models are tested on images from actual farm fields, their accuracy drops to under 30%.

This drop is caused by covariate shift, where the background, lighting, and leaf positioning in the target domain differ significantly from the source domain. Singh et al. [5] introduced the PlantDoc dataset to represent real-world conditions. Researchers have since utilized domain adaptation techniques, such as mixed-domain training and adversarial training, to enforce the extraction of domain-invariant features.

1.2.3 Knowledge-Grounded Support and RAG Systems

A complete agricultural assistant must go beyond classification; it must provide reliable treatment advice to farmers. Traditional chatbots and Large Language Models (LLMs) suffer from hallucinations, generating confident but incorrect treatment advice, which can lead to crop dam-

age if implemented by farmers.

Retrieval-Augmented Generation (RAG), proposed by Lewis et al. [9], resolves this issue by constraining the LLM's response generation to a set of retrieved documents. In agricultural systems, RAG indexes official farming manuals and research guides using vector databases like FAISS [13]. When a user queries the system, relevant paragraphs are retrieved and passed to the LLM to synthesize a grounded, accurate response.

1.3 Motivation

The primary motivation for this project is to address the limitations of existing agricultural classification systems. Standard academic projects often rely on fine-tuning pre-trained networks using transfer learning. While effective, this approach does not demonstrate custom architectural design. Farmers are also hesitant to trust AI predictions without understanding the reasoning behind them. The black-box nature of deep neural networks necessitates explainable AI tools like Grad-CAM [8] to visualize the decision-making process. Finally there is a clear need for a unified platform that not only detects the disease but also offer verified, non-hallucinated treatment guidelines directly sourced from reliable agricultural publications.

1.4 Objectives of the Present Work

The objectives defined for the design and deployment of the proposed system are:

1. To design and implement a custom, lightweight CNN architecture (AgriNet) from scratch in PyTorch.
2. To implement a Mixed-Domain Training pipeline combining laboratory and real-world field datasets to overcome the lab-to-field domain gap.
3. To achieve a validation accuracy exceeding 95% on mixed-domain validation sets.
4. To integrate Grad-CAM to highlight the areas of interest on the crop leaves, making predictions explainable.
5. To construct a local, knowledge-grounded RAG engine using FAISS and Llama-3.1 to retrieve and generate verified treatment recommendations from crop manuals.

6. To develop a web-based portal utilizing Next.js for the UI and FastAPI for backend serving, complete with database persistence for prediction histories.

1.5 Project Outline

The remainder of this report is organized as follows: Chapter 2 describes the dataset characteristics, preprocessing steps, data augmentation techniques, and the class mapping pipeline. Chapter 3 presents the proposed methodology, including the design details of AgriNet, the mathematical formulation of depthwise convolutions, the training loops, and the RAG engine architecture. Chapter 4 illustrates the system design, detailing the DFDs, overall architecture, and database schema. Chapter 5 provides the results analysis, performance comparisons, explainability demonstrations, and concludes the work along with future scopes.

Chapter 2

DATASET PREPARATION AND DOMAIN ADAPTATION

2.1 General

84 A deep learning classifier's capability is fundamentally bound to the quality and diversity of its training data. Exposing neural networks to varied environmental conditions is critical for achieving robust generalization in agricultural computer vision. This chapter details our primary and secondary datasets, the image preprocessing pipeline, data augmentation techniques, and the class mapping logic implemented to fuse laboratory and field imagery.

2.2 Primary Dataset: PlantVillage

53 We utilized the PlantVillage dataset [2] as our primary training source. This corpus contains 54,305 high-resolution, color images of healthy and diseased crop leaves across 14 plant species (e.g., Apple, Corn, Potato, Tomato, and Grape), distributed among 38 biological classes.

Importantly, all PlantVillage photos were captured under controlled laboratory conditions, featuring single leaves detached and laid flat against uniform grey backgrounds under consistent studio lighting. While this simplifies the extraction of clear pathological features, the total lack of background variation limits the network generalisation capability when exposed to unstructured, outdoor conditions.

2.3 Secondary Dataset: PlantDoc

The PlantDoc dataset was added to inject real-world visual noise into the training process. It comprises 2,231 images representing 13 species. Unlike the lab images, PlantDoc photos were taken in actual farms using mobile phones. The pictures show leaves still attached to plants, surrounded by soil, weeds, shifting sunlight, and overlapping branches. Merging this data forces the network to find features that are not dependent on background clutter.

2.4 Preprocessing and Augmentation

80 A strict preprocessing pipeline was applied to all images from both datasets. Every image was resized to 224x224 pixels so it would fit the input requirements of the AgriNet backbone. The

pixel values were scaled between 0 and 1, followed by normalization using standard ImageNet channel statistics.

During the training cycles, data augmentation modified the training split to stop overfitting and boost generalization. The applied transformations included RandomHorizontalFlip to simulate mirrored leaf angles, and RandomRotation to mimic different camera tilts. ColorJitter randomly shifted the brightness, contrast, saturation, and hue to recreate various outdoor lighting scenarios like direct sun or heavy shade.

2.5 Mixed-Domain Dataset Fusion Pipeline

Combining PlantVillage and PlantDoc presented a challenge because they used different naming rules for their class folders. A custom PyTorch dataset class, called MappedPlantDocDataset, was built to fix this issue. It maps the secondary dataset labels directly to the corresponding primary 38-class indices.

A mapping dictionary handled the conversions. It successfully aligned 26 of the 27 PlantDoc classes to their PlantVillage equivalents. One category, ‘Tomato mold leaf’, had to be dropped because it did not have a matching biological class in the primary dataset. Once the mapping was complete, the two datasets were merged using PyTorch’s ConcatDataset function. This fusion created a unified training set containing 56,536 images, as summarized in Table 2.1.

Table 2.1: Dataset Distribution and Class Mapping Summary

Dataset	Image Characteristics	Size (Images)	Classes
PlantVillage	Controlled Lab, Grey Background	54,305	38
PlantDoc	Real-world Field, Noisy Background	2,231	27
Merged Corpus	Mixed-Domain Dataset	56,536	38

2.6 Conclusion

Through the execution of this fusion pipeline, the domain discrepancy between laboratory-captured and field-captured images is addressed. This prepared dataset forms the basis for training the robust classification model described in the following chapter.

Chapter 3

PROPOSED METHODOLOGY

3.1 General

This part covers the architectural choices behind the custom CNN backbone, AgriNet. It breaks down the mathematical logic of the primary building blocks, describes the two-stage training approach, and details how the local Retrieval-Augmented Generation (RAG) system was put together.

3.2 Custom CNN Backbone: AgriNet

Standard 2D convolutions demand substantial computational resources and parameters, hindering deployment on edge devices and mobile phones. To alleviate this bottleneck, we designed AgriNet as a lightweight, resource-efficient alternative. The foundational building blocks of AgriNet are Depthwise Separable Convolutions and Inverted Residual Blocks (AgriBlocks) [6].

3.2.1 Depthwise Separable Convolutions

Depthwise separable convolutions minimize the parametre footprint and multiplication operations by factorizing a standard convolution into two distinct stages: (i) a depthwise convolution, which applies a single spatial filter to each input channel independently, and (ii) a pointwise convolution (1×1), which linearly projects the depthwise outputs into the target channel space.

Let D_K be the spatial dimension of the convolutional kernel, M be the number of input channels, N be the number of output channels, and $D_F \times D_F$ be the spatial dimension of the output feature map. The computational cost of a standard convolution is:

$$\text{Cost}_{\text{standard}} = D_K \cdot D_K \cdot M \cdot N \cdot D_F \cdot D_F \quad (3.1)$$

The computational cost of a depthwise separable convolution is:

$$\text{Cost}_{\text{separable}} = D_K \cdot D_K \cdot M \cdot D_F \cdot D_F + M \cdot N \cdot D_F \cdot D_F \quad (3.2)$$

The ratio of computational reduction is expressed as:

$$\frac{\text{Cost}_{\text{separable}}}{\text{Cost}_{\text{standard}}} = \frac{1}{N} + \frac{1}{D_K^2} \quad (3.3)$$

Assuming a standard kernel size of $D_K = 3$, this decomposition reduces the computational complexity by a factor of 8 to 9 depending on the output channel dimension N , without significant degradation in classification performance.

3.2.2 Inverted Residual Block (AgriBlock)

The AgriBlock serves as the core feature extraction unit of AgriNet, structured as a three-stage block. First, an expansion pointwise convolution (1×1) projects the input to a higher-dimensional space based on the expansion factor t . Second, a depthwise convolution (3×3) extracts spatial features in this high-dimensional space. Third, a projection pointwise convolution (1×1) projects the features back down to the target dimension. To prevent information loss and maintain linear representation, we omit the non-linear activation function after the final projection step, forming a linear bottleneck.

An identity shortcut connection is introduced to sum the block input and output. This skip connection is only applied when the stride equals 1 and the input channels match the output channels. The final network architecture consists of an initial convolutional stem, four sequential AgriBlocks with increasing channel depths, a Global Average Pooling layer, a Dropout layer (configured to 0.3), and a final fully connected classification layer, as outlined in Table 3.1.

Table 3.1: AgriNet Architectural Layout

Layer Type	Input Size	Expansion (t)	Stride
Conv2d Stem	$3 \times 224 \times 224$	-	2
AgriBlock 1	$32 \times 112 \times 112$	4	1
AgriBlock 2	$64 \times 112 \times 112$	6	2
AgriBlock 3	$128 \times 56 \times 56$	6	2
AgriBlock 4	$256 \times 28 \times 28$	6	2
Adaptive Avg Pool	$512 \times 14 \times 14$	-	-
FC Classifier	$512 \times 1 \times 1$	-	-

3.3 Model Training Strategy

The training process was divided into two distinct phases.

3.3.1 Phase 1 Base Model Training Loop

The initial phase trained the model solely on the clean PlantVillage dataset. The process ran for 10 epochs using a batch size of 16. The Adam optimizer handled the weight updates starting with a learning rate of 0.001, while Cross-Entropy served as the loss function. A ReduceLROnPlateau scheduler monitored the validation loss. It automatically cut the learning rate in half whenever the performance stopped improving. The best weights were saved based on the lowest recorded validation loss.

3.3.2 Phase 2 Mixed-Domain Fine-Tuning Loop

The second phase loaded the saved weights from the first phase and switched the dataset to the merged PlantVillage and PlantDoc collection. To avoid catastrophic forgetting, the learning rate was dropped to 0.0001. The model underwent another 10 epochs of fine-tuning. The final weights were saved for API deployment.

3.4 Knowledge-Grounded RAG Chat Engine

A Retrieval-Augmented Generation pipeline was built to supply verified agricultural treatments.

3.4.1 Document Ingestion and Chunking Strategy

We developed an automated pipeline to ingest official agricultural publications (e.g., Tomato [19], Potato [20], Maize [21], Apple [22]) and Integrated Disease Management manuals [23].

The system extracts raw text from these PDFs and processes them using LangChain's RecursiveCharacterTextSplitter.

The chunk size is configured to 1,000 characters with a 200-character sliding overlap. This overlap maintains semantic continuity across boundary points.

3.4.2 Vector Database and Retrieval Mechanics

To perform fast similarity search, we embed the text chunks into dense 384-dimensional vectors using a Sentence-Transformers model. These embeddings are stored locally in a Facebook AI

Similarity Search (FAISS) vector index. When a user submits an inquiry, the query is mapped to the same embedding space, and the system runs a cosine-similarity search to retrieve the top $k = 3$ most relevant document chunks.

3.4.3 Context-Grounded Generation (Groq Llama-3.1)

The retrieved document chunks are formatted alongside the user query within a strictly constrained system prompt. This prompt is transmitted to the llama-3.1-8b-instant model via the Groq API. The model is explicitly instructed to synthesize answers solely from the retrieved context; if the context lacks relevant information, the model states that it cannot answer. This setup ensures that all responses are context-grounded and contain references to the source documentation, minimizing hallucinations.

3.5 Conclusion

By combining a lightweight CNN classifier with a localized RAG engine, the proposed methodology addresses both disease identification and treatment advisory within a single, integrated architecture.

Chapter 4

SYSTEM DESIGN AND IMPLEMENTATION

4.1 General

This chapter delineates the architectural design and structural implementation details of the AgriVision application, including data flow, database models, and hardware specifications.

4.2 Data Flow Diagrams (DFDs)

Data Flow Diagrams (DFDs) track the progression and transformations of data throughout the system.

At the Context Level (Level 0), the system acts as a monolithic interface connecting the user, the database, and the vector index. Farmers upload leaf images and input text queries. The system outputs classification predictions, Grad-CAM overlays, RAG responses, and transaction logs.

The Level 1 DFD decomposes the system into three primary processes: (i) user authentication, which verifies credentials and issues JWTs; (ii) image prediction, which processes uploads through AgriNet, computes Grad-CAM heatmaps, and saves logs to PostgreSQL; and (iii) the RAG chat loop, which routes user queries through the FAISS vector database and LLM APIs. Figure 4.1 illustrates these DFD levels.

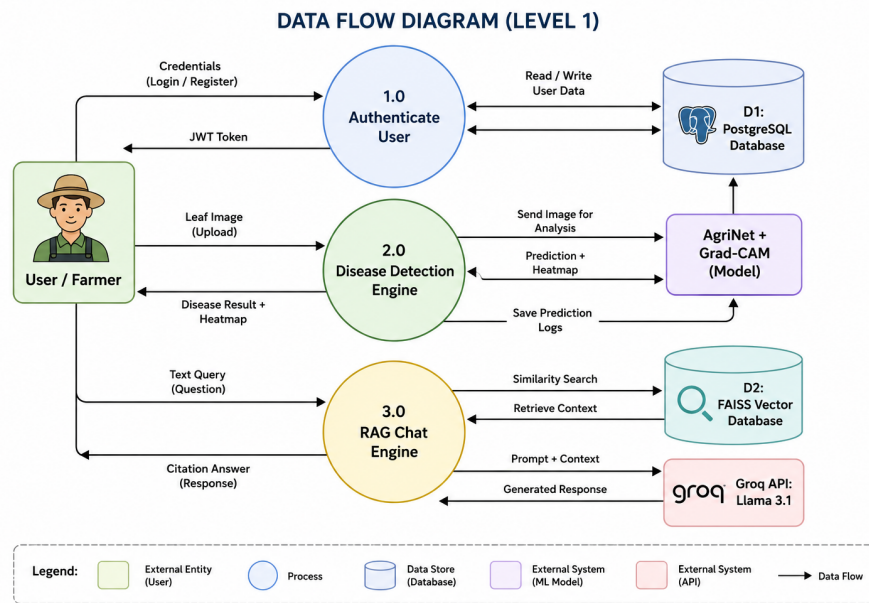


Figure 4.1: System Data Flow Diagram (DFD Level 0 and Level 1)

4.3 Project Architecture

The system utilizes a decoupled client-server architecture to ensure high performance, security, and scalability. The separation of concerns between the presentation layer, the application logic, and the machine learning/retrieval subsystems is illustrated in the System Architecture Diagram in Figure 4.2.

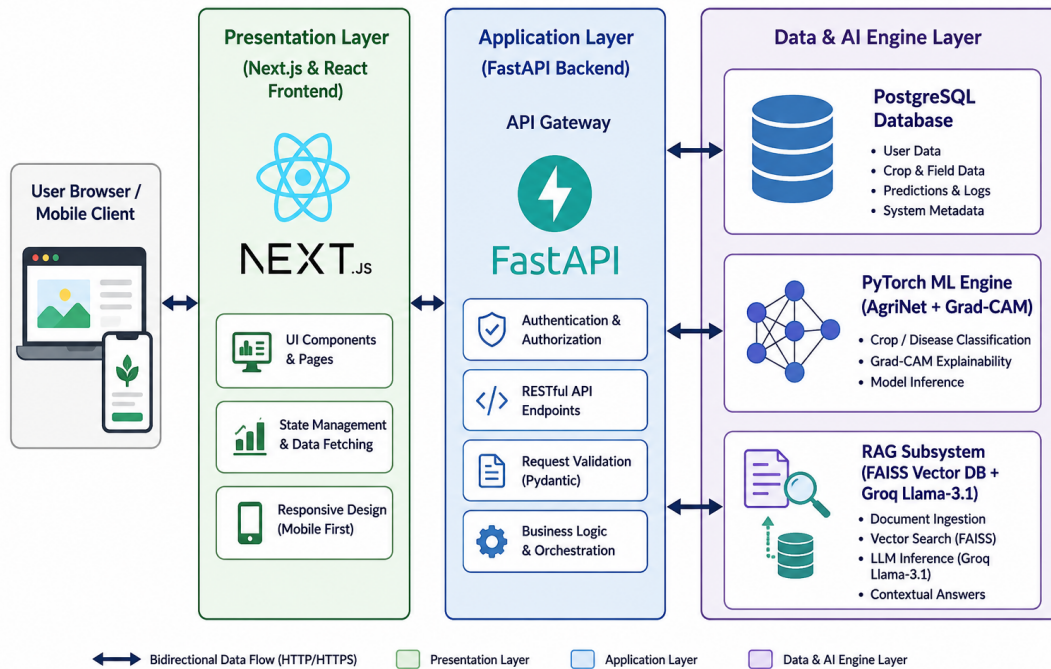


Figure 4.2: AgriVision Multi-Layer System Architecture Diagram.

The architecture is divided into three primary layers:

- 1. Presentation Layer (Frontend):** Built using Next.js 14 and React. It serves as a pure client and manages the presentation layer using custom vanilla CSS for glassmorphic visuals, with Framer Motion handling smooth layout transitions. It communicates with the backend via REST APIs for authentication, prediction requests, and conversational chat queries.
- 2. Application Logic Layer (Backend):** Powered by FastAPI on Python 3.11. This layer acts as the central orchestrator. It manages API routing, handles secure user authentication using JSON Web Tokens (JWT) and password hashing with Bcrypt, manages request/response payloads, and coordinates services between the machine learning models and the database.
- 3. Data and AI Engine Layer:** Consists of three core sub-components:
 - **PostgreSQL Database:** Persistent storage for user records, prediction histories, and diagnostic statistics, integrated directly with the backend using the Python Prisma Client (prisma-client-py).

- **Deep Learning Classifier (AgriNet + Grad-CAM):** Executes the lightweight PyTorch model for disease classification and generates explainability heatmaps using Grad-CAM.
- **RAG Chat Subsystem:** Connects the FAISS vector database containing the indexed agricultural manuals with the Groq API (llama-3.1-8b-instant model) to serve grounded, verified treatment advice.

4.4 Database and Schema Design

The relational database is configured with two primary models to track users and prediction logs. The specific structure and data types for these tables are detailed in Table 4.1 and Table 4.2.

- **User Table:** Stores the unique ID (cuid), user's name, email address (marked unique), Bcrypt-hashed password, and the account creation timestamp.
- **Prediction Table:** Tracks crop scans and holds a unique prediction ID, a foreign key referencing the User ID (with cascade delete enabled), the identified plant name, the predicted disease name, the model's confidence score, a boolean flag indicating if the plant is healthy, a serialized JSON array representing the top-5 classification probabilities, and a serialized JSON object containing the treatment recommendations. Each record includes a timestamp of when the scan occurred.

Table 4.1: User Table Schema

Field	Data Type	Constraints	Description
id	String	Primary Key (cuid())	Unique identifier for the user
name	String	-	Full name of the user
email	String	Unique	Email address (used for login)
password	String	-	Bcrypt hashed password string
createdAt	DateTime	Default (now())	Timestamp of account registration

Table 4.2: Prediction Table Schema

Field	Data Type	Constraints	Description
id	String	Primary Key (cuid())	Unique identifier for prediction record
userId	String	Foreign Key (User.id)	ID of the user who scanned the leaf
plantName	String	-	Name of the plant type (e.g., Tomato)
diseaseName	String	-	Detected disease or healthy state
confidence	Float	-	Confidence score (percentage value)
isHealthy	Boolean	-	True if healthy, False if diseased
topPredictions	String	Default ([])	Serialized JSON string of top-5 predictions
recommendations	String	Default ({})	Serialized JSON string of treatment options
createdAt	DateTime	Default (now())	Timestamp of when prediction was logged

4.5 Tech Stack and Requirements

The core languages used include Python 3.11 for the backend and TypeScript for the frontend UI. Key libraries involve PyTorch, Torchvision, LangChain, FAISS, Sentence-Transformers, FastAPI, and Prisma. The database runs on PostgreSQL 15 or newer.

The minimum hardware needed to run inference includes an Intel i5 CPU, 8 GB of RAM, and an NVIDIA GTX 1050 Ti. For training the model, an Intel i7 or Ryzen 7 paired with an RTX 3060 and 16 GB of RAM is recommended. The actual training was completed on a local setup using an NVIDIA GeForce GTX 1650 with 4GB of VRAM.

4.6 User Interface Design

To provide an intuitive and accessible experience for farmers and researchers, the AgriVision web application features a responsive user interface designed with a clean, glassmorphic dark theme. The web portal includes several primary pages and panels:

4.6.1 Plant Disease Diagnosis Page

The main interface allows users to upload leaf images or use their device camera to capture real-time plant leaves. Upon uploading, the backend processes the image using the AgriNet model

and returns the prediction along with the classification confidence and a Grad-CAM heatmap overlay to explain the model's focus. Figure 4.3 displays the plant disease identification panel with the Grad-CAM visualization.

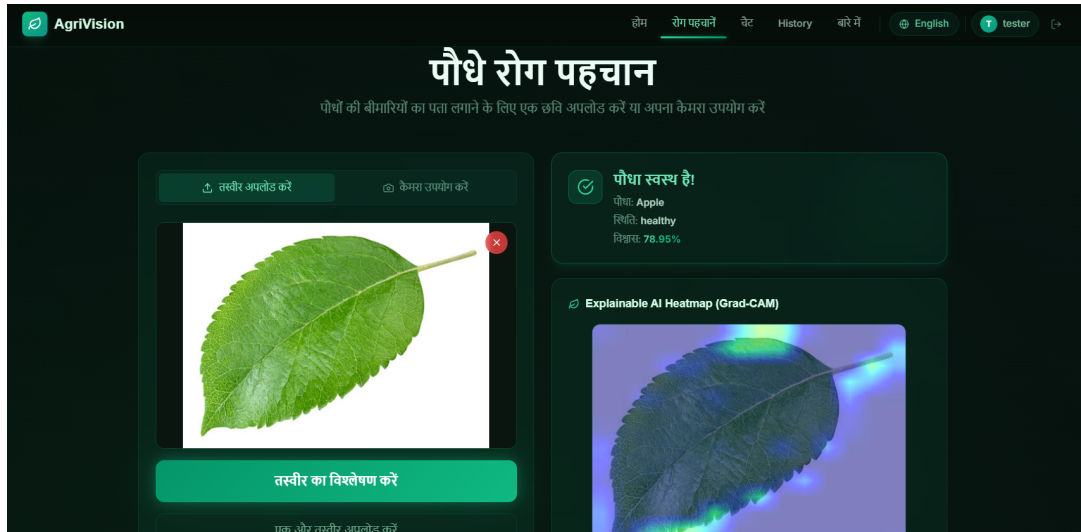


Figure 4.3: AgriVision Diagnosis Interface showing predictions and Grad-CAM explanation.

4.6.2 AI Chat Assistant Panel

The application integrates a Retrieval-Augmented Generation (RAG) chat helper called ‘AI Assistant’ (AI). The chatbot utilizes the user query and the classification context to retrieve validated treatments from the local FAISS index and generate natural-language recommendations supporting multilingual output (e.g., Hindi). Figure 4.4 illustrates the conversational interface assisting the user.

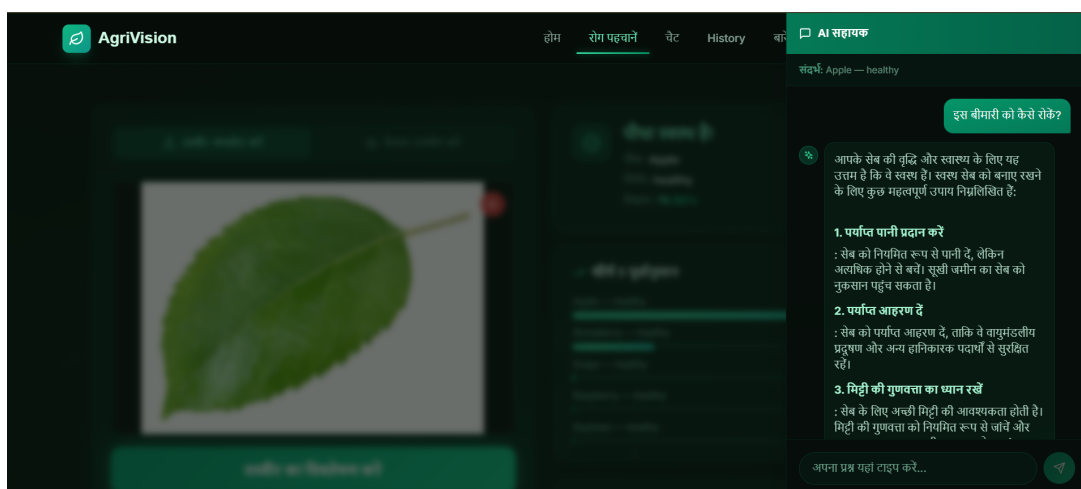


Figure 4.4: Interactive AI Chat Assistant panel helping the user.

4.6.3 Diagnosis History Dashboard

To keep track of previous scans, the web portal provides a dedicated history dashboard showing a chronological list of prior diagnoses, including crop species, identified disease, prediction timestamp, and confidence. Users can click on any past record to open details or download reports. Figure 4.5 shows the history dashboard.

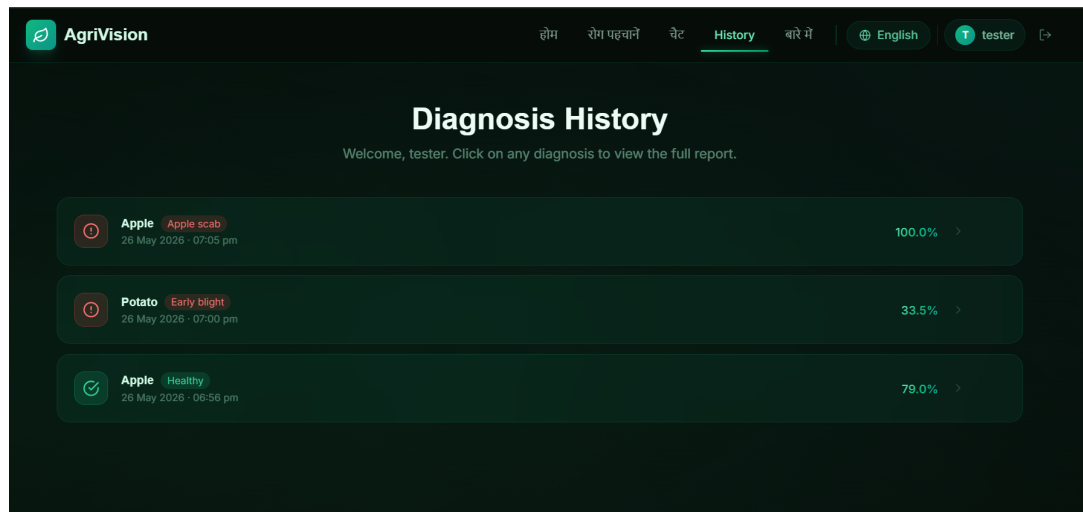


Figure 4.5: Chronological Diagnosis History Dashboard listing previous predictions.

4.7 Conclusion

The system design provides a scalable architecture where frontend interactions, heavy deep learning inference, and database logs are handled by decoupled, dedicated services.

Chapter 5

RESULT ANALYSIS AND DISCUSSION

5.1 Experimental Results

We systematically tracked the performance of the custom AgriNet model during both the initial base training phase and the subsequent mixed-domain fine-tuning stage.

5.1.1 Phase 1 Base Model Training Results

During the initial phase on the pristine PlantVillage corpus, the network exhibited rapid convergence, reaching a validation accuracy of 94.68% by the second epoch. By the final epoch (10), the model attained its lowest validation loss of 0.0396 and a peak validation accuracy of 98.75%. These results indicate that the Inverted Residual Blocks effectively extract core leaf features when background noise is minimized. The detailed epoch-by-epoch training progress is reported in Table 5.1.

Table 5.1: Phase 1 Training Metrics

Epoch	Train Loss	Train Acc	Val Loss	Val Acc
1	0.8001	76.35%	0.3566	89.08%
2	0.2985	90.46%	0.1645	94.68%
3	0.2098	93.30%	0.1371	95.51%
4	0.1691	94.57%	0.1120	96.27%
5	0.1445	95.33%	0.0783	97.50%
6	0.1214	96.09%	0.0636	97.81%
7	0.1120	96.42%	0.0883	97.25%
8	0.1044	96.57%	0.0513	98.23%
9	0.0939	96.98%	0.0964	96.85%
10	0.0853	97.21%	0.0396	98.75%

5.1.2 Phase 2 Mixed-Domain Fine-Tuning Results

Phase 2 introduced the challenging, noisy PlantDoc images into the training pipeline. Although natural backgrounds, occlusion, and variable illumination increased training difficulty, the model converged within 9 epochs, achieving a mixed-domain validation accuracy of 96.14%. This validates the effectiveness of our domain adaptation approach in handling natural field environments. Table 5.2 provides the fine-tuning training metrics.

Table 5.2: Phase 2 Fine-Tuning Metrics

Epoch	Train Loss	Train Acc	Val Loss	Val Acc
1	0.5123	90.70%	0.3257	93.83%
2	0.3337	92.69%	0.2591	94.59%
3	0.2831	93.48%	0.2231	95.19%
4	0.2525	93.89%	0.2065	95.30%
5	0.2349	94.14%	0.1838	95.68%
6	0.2247	94.34%	0.1782	95.95%
7	0.2101	94.62%	0.1774	95.98%
8	0.2035	94.67%	0.1778	95.71%
9	0.1936	94.99%	0.1587	96.14%
10	0.1877	94.96%	0.1652	96.01%

5.2 Performance Evaluation Metrics

We evaluated classification performance using standard statistical metrics: Precision, Recall, and F1-score. Precision measures the ratio of true positive predictions to all predicted positives; Recall measures the ratio of true positives to all actual positive instances; and the F1-score is computed as their harmonic mean. On the combined validation dataset, the model achieved a weighted average of 0.99 across all three metrics. The corresponding 38-class confusion matrix is illustrated in Figure 5.1.

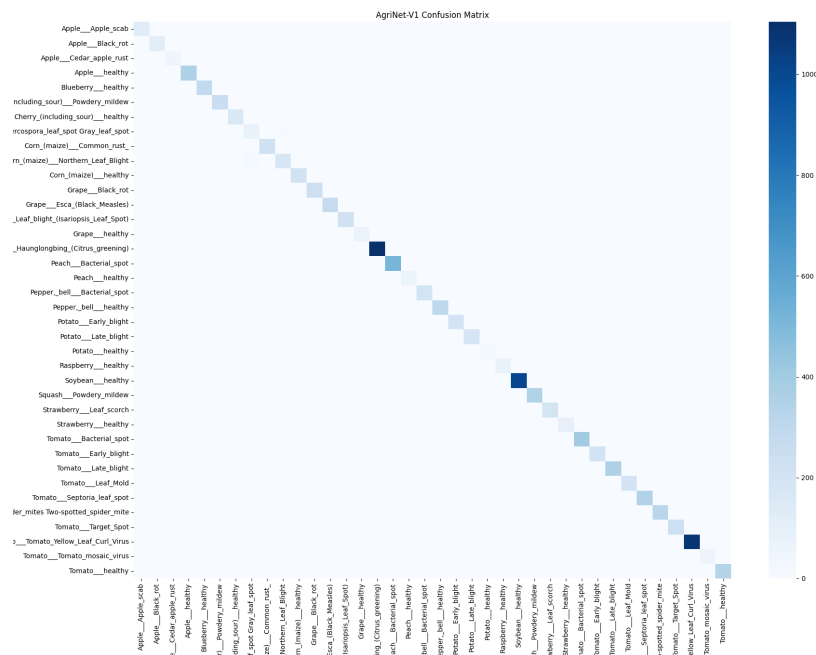


Figure 5.1: Confusion Matrix for 38-Class Mixed-Domain Predictions

5.3 Zero-Shot Field Testing

To validate the real-world robustness of the proposed system, zero-shot field testing was conducted on entirely unseen images from local farms. Despite the presence of complex backgrounds and varying illumination, the model correctly identified the diseases, demonstrating strong generalization beyond the training distribution, with a sample prediction shown in Figure 5.2.

Zero-Shot Domain Transfer: Testing on Wild Field Data (PlantDoc)

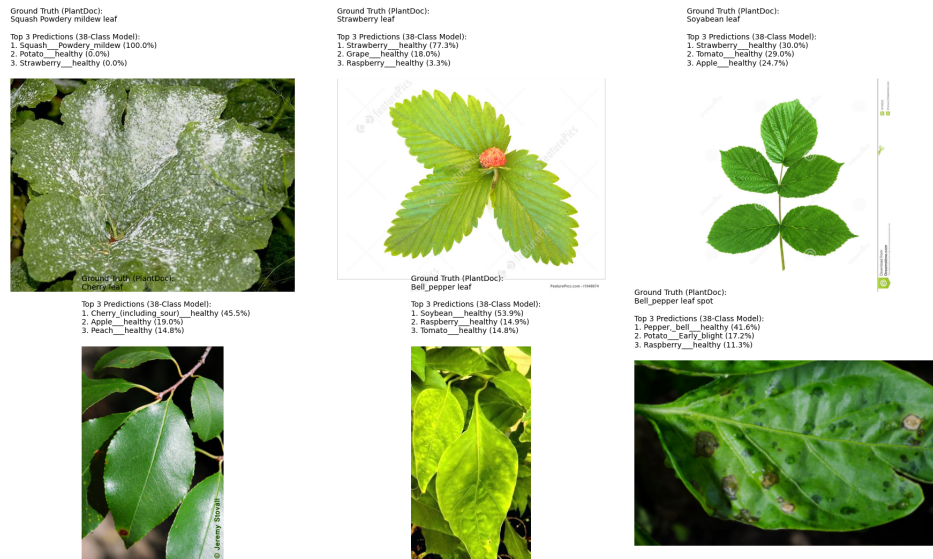


Figure 5.2: Zero-Shot Field Test Prediction on Unseen Data

5.4 Explainable AI (XAI) via Grad-CAM Visualizations

Gradient-weighted Class Activation Mapping (Grad-CAM) was added to ensure the predictions are transparent. Grad-CAM looks at the gradients of the target class score relative to the feature map activations in the final convolutional layer. It globally pools these gradients to figure out the importance of each feature map. Applying a ReLU activation isolates the specific features that push the model toward the target class:

$$L_{\text{Grad-CAM}}^c = \text{ReLU} \left(\sum_k \alpha_k^c A^k \right) \quad (5.1)$$

When this resulting heatmap is laid over the original image, it shows the model actively focusing on leaf lesions rather than background soil or shadows, as demonstrated by the overlay heatmap in Figure 5.3.

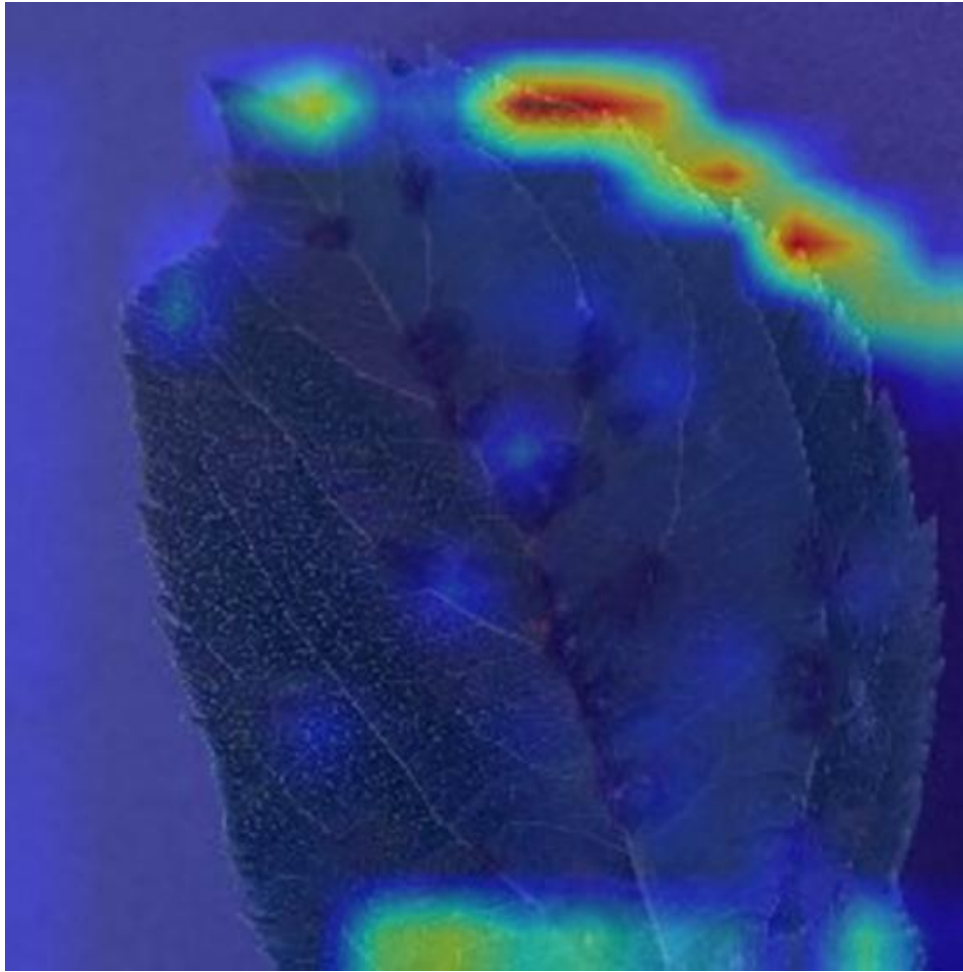


Figure 5.3: Grad-CAM Heatmap Overlaid on Diseased Leaf

5.5 Conclusion and Future Work

5.5.1 Summary of Findings

The proposed application proves that a custom, lightweight CNN can deliver high accuracy while using minimal computational resources. The mixed-domain training approach successfully lowered the model's sensitivity to background noise, resulting in a 96.14% accuracy rate on field images. By adding a local RAG engine, the platform guarantees that farmers receive accurate treatment advice pulled straight from official crop manuals.

5.5.2 Future Scope

Several areas remain open for future development. Converting the AgriNet weights into ONNX or TensorFlow Lite formats would allow the model to run offline on mobile phones. The network architecture could be modified to support dual-task learning, adding a regression head

to calculate the percentage of leaf area affected by the disease. Interfacing the backend with drone imagery would enable large-scale farm monitoring. Finally, translating the user interface and the RAG outputs into local regional languages would greatly improve accessibility for rural farming communities.

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